SET A - QUESTION 1 (20 marks)

a) Diagram 1 shows a parallelogram ABCD drawn on a Cartesian plane.

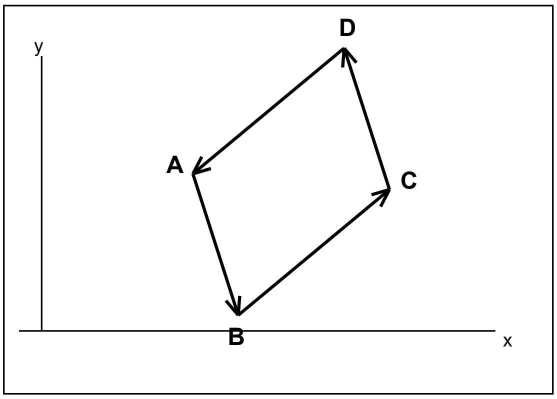

Diagram 1: Parallelogram ABCD

It is given that AB = 5i - 2j and BC = 3i + 6j. Find AC.

b) The following information refers to the vectors m and n.

m = (6 , 3k-2)   n = (9 , 6)

It is given that m = hn, where m is parallel to n and h is a constant. Calculate the value of,

i. h

ii. k

c) i. It is given that PQ = 3i - 4j and PR = 8i + 2j. Find,

(a) QR

(b) the unit vector in the direction of QR.

ii. Given XY = 4i + tj, where t is a constant and XY is parallel to ZW = 12i - 9j. Find the value of t.

d) In Diagram 2, EFGH is a quadrilateral, diagonals EG and FH intersect at point Y.

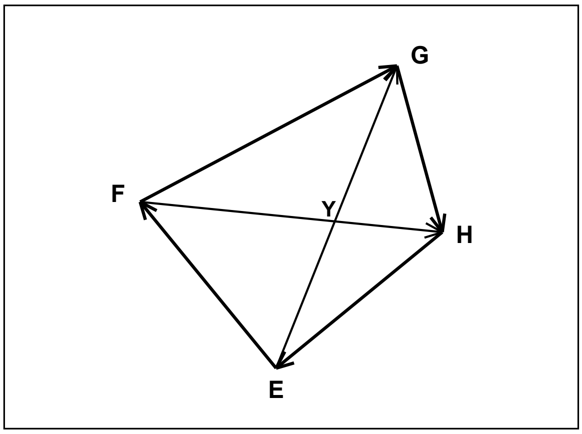

Diagram 2: Quadrilateral EFGH

It is given that EY : YG = 1 : 3, EF = 8a, EH = 12b, and FG = 16a + 12b.

i. Express in terms of a and b

(a) FH

(b) EY

ii. Find the ratio for FY : YH.

[Total: 20 marks]

SET B - QUESTION 1 (20 marks)

a) Diagram 1 shows a parallelogram JKLM drawn on a Cartesian plane.

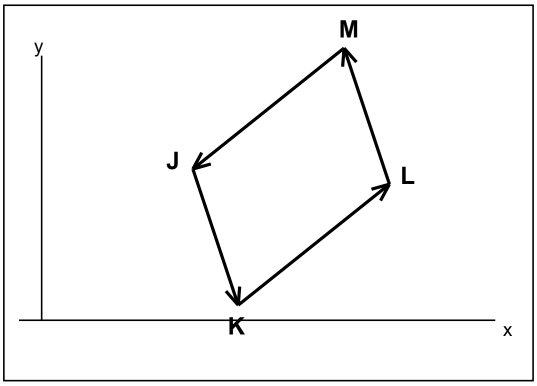

Diagram 1: Parallelogram JKLM

It is given that JK = 2i + 7j and KL = 6i - j. Find JL.

b) The following information refers to the vectors u and v.

u = (4h , 10) v = (5 , 2k+1) (column vectors)

It is given that u = 4v, where u is parallel to v. Calculate the value of,

i. h

ii. k

c) i. It is given that ST = 6i - 3j and SU = 2i + 9j. Find,

(a) TU

(b) the unit vector in the direction of TU.

iii. Given CD = 15i - 6mj, where m is a constant and CD is parallel to EF = 5i - 4j. Find the value of m.

d) In Diagram 2, NOPQ is a quadrilateral, diagonals NP and OQ intersect at point Z.

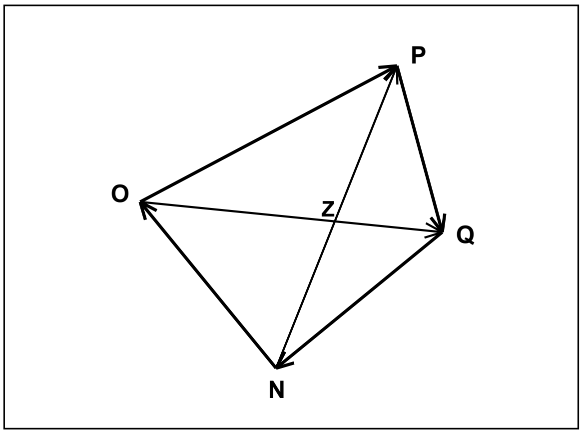

Diagram 2: Quadrilateral NOPQ

It is given that NZ : ZP = 2 : 3, NO = 10c, NQ = 15d, and OP = 5c + 15d.

i. Express in terms of c and d

(a) OQ

(b) NZ

ii. Find the ratio for OZ : ZQ. [4 marks]

[Total: 20 marks]